# 이 경 은 Kyungeun Lee

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## CURRENT POSITION

**Assistant Professor** 2025.9 –
Department of Artificial Intelligence, Inha University PRESENT

Research Focus:
Information-theoretic analysis of learned representations
Development of tabular prediction models for real-world applications
Multimodal learning with tabular data and other data modalities

## EDUCATION

**Seoul National University**, Seoul, South Korea 2016.9 – 2023.2
Ph.D. in Department of Intelligence and Information
(Advisor: Wonjong Rhee)
(Thesis: Understanding Mutual Information in Contrastive Representation Learning)

**Seoul National University**, Seoul, South Korea 2010.3 – 2014.2
Bachelor in Civil and Environmental Engineering

**Kangwon Science High School** 2008.3 – 2010.2
Early graduation

## CAREERS

**LG AI Research**, Seoul, South Korea 2023.1 – 2025.8
AI Researcher, Data Intelligence Lab.

**Korea Cyber University/Graduate School**, Seoul, South Korea 2024.3 – 2025.2
Visiting professor (Deep Learning, Advanced AI)

**Samsung Engineering**, Seoul, South Korea 2014.3 – 2015.3
Supervisor engineer

## PROJECT EXPERIENCE

**Demand response pilot program design and evaluation for 31,615 customers**, *Seoul National University, Korean Power Exchange, LG Uplus, Encored technologies* 2016 – 2018

**UTM traffic management system design and construction in low altitude**, *Seoul National University, KAIST* 2017 – 2019

**Empirical investigation of the statistical characteristics of deep representations**, *Seoul National University* 2018 – 2019

**Information-theoretic analysis of deep representations**, *Seoul National University* 2020 – 2022

**3D augmentation method for object detection in point-cloud datasets**, *Seoul National University* 2022

**Anomaly detection and tabular prediction for an automated manufacturing system**, *LG AI Research* 2023 – 2025

## PUBLICATIONS *(Sorted by date)*

### List of papers as **first author**

[1] Kyungeun Lee et al., Range-limited Augmentation for Few-shot Learning in Tabular Data, 31st SIGKDD Conference on Knowledge Discovery and Data Mining - Research track, 2025.

[2] Kyungeun Lee et al., A Benchmark Suite for Evaluating Neural Mutual Information Estimators on Unstructured Datasets, The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.

[3] Kyungeun Lee et al., Towards a Rigorous Analysis of Mutual Information in Contrastive Learning, Neural Networks, 2024. (SCIE, IF: 6.0)

[4] Kyungeun Lee et al., Binning as a Pretext Task: Improving Self-Supervised Learning in Tabular Domains, The Forty-First International Conference on Machine Learning (ICML), 2024.

[5] Kyungeun Lee et al., DDP-GCN: Multi-graph Convolutional Network for Spatiotemporal Traffic Forecasting, Transportation Research Part C: Emerging Technologies, 2022. (SCIE, IF: 7.6)

*Top 2% of citations and captures among the papers published in the same year*

[6] Kyungeun Lee et al., Statistical Characteristics of Deep Representations: An Empirical Investigation, International Conference on Artificial Neural Networks, 2021.

[7] Kyungeun Lee et al., Short-term Traffic Prediction with Deep Neural Networks: A Survey, IEEE Access, 2021. (SCIE, IF: 3.4)

[8] Kyungeun Lee et al., Assuring Explainability on Demand Response Targeting via Credit Scoring, Energy, 2018. (SCIE, IF: 9.0)

### List of papers as **co-author**

[1] S Yoon, S Yoon, H Lee, Y Sim, S Choi, **K Lee**, H Cho, W Lim, Diffusion based Semantic Outlier Generation via Nuisance Awareness for Out-of-Distribution Detection, The 39th Annual AAAI Conference on Artificial Intelligence (AAAI), 2025.

[2] M Eo, **K Lee**, H Cho, D Kim, Y Sim, W Lim, Representation Space Augmentation for Effective Self-Supervised Learning on Tabular Data, The 39th Annual AAAI Conference on Artificial Intelligence (AAAI), 2025.

[3] J Shin, J Kim, **K Lee**, H Cho, W Rhee, Diversified and Realistic 3D Augmentation via Iterative Construction, Random Placement, and HPR Occlusion, The 37th Annual AAAI Conference on Artificial Intelligence (AAAI), 2023.

[4] E Lee, **K Lee**, H Lee, E Kim, W Rhee, Defining virtual control group to improve customer baseline load calculation of residential demand response, Applied Energy, 2019. (SCIE, 10.1)

## TEACHING

**Inha University, Assistant professor** 2025

- 2025 Fall: Natural Language Processing, Probability Theory
- 2026 Spring: Machine Learning, Artificial Intelligence, Deep Learning (Graduate)

**Korea Cyber University/Graduate School, Visiting professor** 2024 – 2025

- Deep Learning, Advanced Artificial Intelligence (Graduate)

**LG AI Research, Mentor** 2023 – 2025

- Research mentoring on tabular prediction for quality forecasting (LG Innotek, LG Display)

**Seoul National University, TA** 2016 – 2021

- Machine Learning, Deep Learning, Information Theory

**Seoul National University Cooperation R&D Center, Samsung, Data Science Class TA** 2019

## INVITED LECTURES & OUTREACH ACTIVITIES

**Tabular Learning and Prediction (Invited Lecture),** Inha University 2025

**Tabular Learning (Industry Invited Lecture),** LG Display 2025

**Understanding and Applications of Artificial Intelligence (Outreach Lecture),** Yeouido High School 2025

**Tabular Learning (Invited Lecture),** Korea Cyber University 2023

**What is Artificial Intelligence? (Theory and Hands-on Practice) (Outreach Lecture),** Wooseok Middle School 2018

## AWARDS & SCHOLARSHIPS

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|---|---|
| **LG Awards**, LG | 2024 |
| **Yulchon AI Young Researcher Scholarship,** Seoul National University | 2022 |
| **Teaching Assistants Scholarship,** Seoul National University | 2021 |
| **Merit-Based Scholarship**, Seoul National University | 2019 |
| **Daerim Suam Scholarship**, Daerim Suam Scholarship Foundation | 2011 – 2013 |
| **Samsung SDS Scholarship**, Samsung SDS | 2009 – 2013 |
| **Merit-Based Scholarship**, Seoul National University | 2010 |